



VIT[®]
Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)

School of Computer Science Engineering and Information Systems
Winter Semester 2025-26

Agile Development Process and DevOps Lab ISWE406P

LAB FAT

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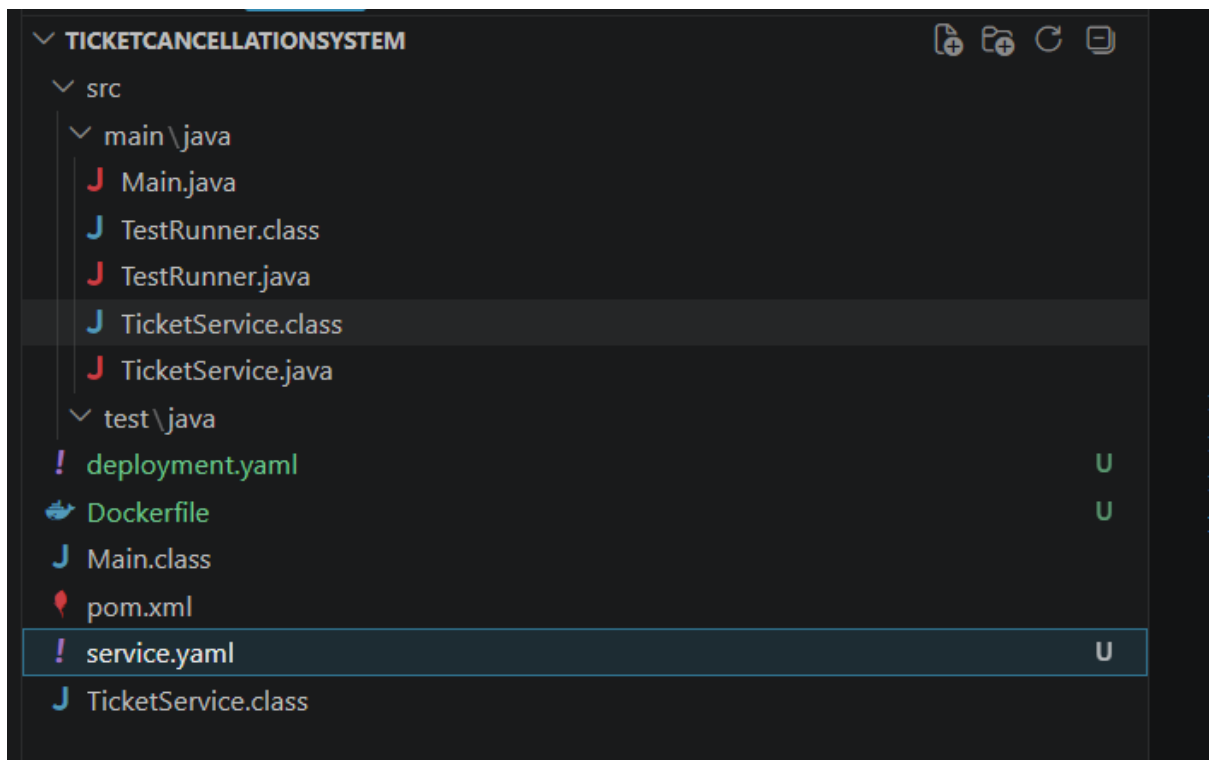
Slot: L47+ L48

QUESTION:

Online Ticket Cancellation System

TOOLS USED

- Java-vs code
- JUnit
- Jenkins
- Docker
- Kubernetes
- GitHub



The screenshot shows an IDE with the following tabs: TicketService.java, Main.java, pom.xml, TicketService.class, deployment.yaml U, service.yaml U, Dockerfile U, and Test.java. The Main.java file is open, showing the following code:

```
src > main > java > J Main.java > Main > main(String[])
1 | package src.main.java;
2 | public class Main {
3 |
4 |     Run | Debug
5 |     public static void main(String[] args) {
6 |         TicketService ts = new TicketService();
7 |         double r1 = ts.refund(30, 1000);
8 |         double r2 = ts.refund(15, 500);
9 |         double r3 = ts.refund(5, 0);
10 |
11 |         System.out.println("Refund (30 hrs): " + r1);
12 |         System.out.println("Refund (15 hrs): " + r2);
13 |         System.out.println("Refund (5 hrs): " + r3);
14 |     }
15 | }
```

A tooltip for the `println` method is displayed, showing its signature `void java.io.PrintStream.println(String x)` and a description: "Prints a String and then terminates the line. This method behaves as though it invokes `print(String)` and then `println()`." The tooltip also lists the parameter `x` as "The String to be printed." and the source as "Source: Java 21.0.9 (module: java.base)".

The bottom panel shows the PROBLEMS, OUTPUT, DEBUG CONSOLE, TERMINAL, and PORTS tabs. The OUTPUT tab is active, showing the following output:

```
PS C:\Users\Wadhumitha\Desktop\TicketCancellationSystem> javac TicketService.java Main.java
PS C:\Users\Wadhumitha\Desktop\TicketCancellationSystem> java Main
Refund (30 hrs): 1000.0
Refund (15 hrs): 500.0
Refund (5 hrs): 0.0
PS C:\Users\Wadhumitha\Desktop\TicketCancellationSystem>
```

The bottom status bar shows the file is at Line 14, Column 2, with 4 spaces, UTF-8 encoding, CRLF line endings, and the Java language.

The screenshot shows an IDE with the following tabs: TicketService.java, Main.java, pom.xml, TicketService.class, deployment.yaml U, service.yaml U, Dockerfile U, and Test.java. The TicketService.java file is open, showing the following code:

```
src > main > java > J TicketService.java > TicketService > cancelTicket(double, int)
2 | public class TicketService {
3 |
4 |     public double cancelTicket(double ticketPrice, int hoursBeforeDeparture) {
5 |
6 |         if (hoursBeforeDeparture < 12) {
7 |             return ticketPrice * 0.5; // 50% refund
8 |         } else if (hoursBeforeDeparture >= 12) {
9 |             return ticketPrice * 0.5; // 50% refund
10 |         } else {
11 |             return 0; // no refund
12 |         }
13 |     }
14 |
15 | }
16 |
17 |
18 |
19 | }
```

The left sidebar shows the Explorer view with the following structure:

- src
 - main.java
 - Main.java
 - TestRunner.class
 - TestRunner.java
 - TicketService.class
 - TicketService.java
- test.java
- deployment.yaml U
- Dockerfile U
- Main.class
- pom.xml
- service.yaml U
- TicketService.class

```
! deployment.yaml
1  apiVersion: apps/v1
2  kind: Deployment
3  metadata:
4    name: ticket-app
5  spec:
6    replicas: 1
7    selector:
8      matchLabels:
9        app: ticket
10   template:
11     metadata:
12       labels:
13         app: ticket
14     spec:
15       containers:
16       - name: ticket-container
17         image: nginx
18         ports:
19       - containerPort: 80
```

```
Dockerfile > ...
1  FROM eclipse-temurin:17-jdk
2
3  WORKDIR /app
4
5  COPY src/main/java/ /app/
6
7  RUN javac TicketService.java TestRunner.java
8
9  CMD ["java", "TestRunner"]
```

```
pom.xml
1  <project xmlns="http://maven.apache.org/POM/4.0.0"
22  <build>
23  <plugins>
24  <!-- Maven Surefire Plugin -->
25  <plugin>
26  <groupId>org.apache.maven.plugins</groupId>
27  <artifactId>maven-surefire-plugin</artifactId>
28  <version>3.0.0</version>
29  </plugin>
30  </plugins>
31 </build>
32
33 </project>
```

```
! service.yaml
1  apiVersion: v1
2  kind: Service
3  metadata:
4  | name: ticket-service
5  spec:
6  | type: NodePort
7  | selector:
8  | | app: ticket
9  | ports:
10 | - protocol: TCP
11 |   port: 80
12 |   targetPort: 8080
13 |   nodePort: 30007
```

```
PS C:\javac TicketService.java TestRunner.java in System\src\main\java>
PS C:\Users\Madhumitha\Desktop\TicketCancellationSystem\src\main\java> java TestRunner
TestFullRefund: PASSED
TestHalfRefund: PASSED
TestNoRefund: PASSED
TestInvalidInput: PASSED

Total Tests: 4
Passed: 4
Failed: 0
PS C:\Users\Madhumitha\Desktop\TicketCancellationSystem\src\main\java>
```

```
PS C:\Users\Madhumitha\Desktop\TicketCancellationSystem> git init
Initialized empty Git repository in C:\Users\Madhumitha\Desktop\TicketCancellationSystem\.git\
PS C:\Users\Madhumitha\Desktop\TicketCancellationSystem> git add .
PS C:\Users\Madhumitha\Desktop\TicketCancellationSystem> git commit -m "Initial commit"
[master (root-commit) e22fc28] Initial commit
9 files changed, 145 insertions(+)
create mode 100644 Main.class
create mode 100644 TicketService.class
create mode 100644 pom.xml
create mode 100644 src/main/java/Main.java
create mode 100644 src/main/java/TestRunner.class
create mode 100644 src/main/java/TestRunner.java
create mode 100644 src/main/java/TicketService.class
create mode 100644 src/main/java/TicketServiceTest.java
PS C:\Users\Madhumitha\Desktop\TicketCancellationSystem> git branch -M main
PS C:\Users\Madhumitha\Desktop\TicketCancellationSystem> git remote add origin https://github.com/madhumitha15-bot/ticket-cancellation-system.git
PS C:\Users\Madhumitha\Desktop\TicketCancellationSystem> git push -u origin main
Enumerating objects: 13, done.
Counting objects: 100% (13/13), done.
Delta compression using up to 12 threads
Compressing objects: 100% (11/11), done.
Writing objects: 100% (13/13), 3.49 KiB | 1.16 MiB/s, done.
Total 13 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), done.
To https://github.com/madhumitha15-bot/ticket-cancellation-system.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
PS C:\Users\Madhumitha\Desktop\TicketCancellationSystem>
```

Console - #1 - Jenkins

madhumitha15-bot/ticket-canc...

github.com/madhumitha15-bot/ticket-cancellation-system

madhumitha15-bot / ticket-cancellation-system

Type to search

<> Code

Issues

Pull requests

Agents

Actions

Projects

Wiki

Security and quality

Insights

Settings

On April 24 we'll start using GitHub Copilot interaction data for AI model training unless you opt out. [Review this update](#) and manage your preferences in your [GitHub account settings](#).

ticket-cancellation-system

Public

Pin

Watch 0

Fork 0

main 1 Branch 0 Tags

Go to file

Add file

<> Code

madhumitha15-bot Initial commit e22fc28 · 24 minutes ago 1 Commit

src/main/java Initial commit 24 minutes ago

Main.class Initial commit 24 minutes ago

TicketService.class Initial commit 24 minutes ago

pom.xml Initial commit 24 minutes ago

README

About

No description provided.

Activity

0 stars

0 watching

0 forks

Releases

No releases published

Create a new release

Packages

Jenkins pipeline

localhost:8080/view/all/newJob



Jenkins



All



New Item



New Item

Enter an item name

TicketPipeline

Select an item type



Pipeline

Build, test, and deploy using pipelines. Supports stages, parallel work, and running on multiple agents.



Freestyle project

Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.



Multi-configuration project

Suitable for projects that need a large number of different configurations, such as testing on multiple environments, platform-specific builds, etc.



Folder

Creates a container that stores nested items in it. Useful for grouping things together. Unlike view, which is just a filter, a folder creates a separate namespace, so you can have multiple things of the same name as long as they are in different folders.



Organization Folder

OK

Activate Windows

Go to Settings to activate Windows.



Jenkins



TicketPipeline



Configure



Configure



General



Triggers



Pipeline



Advanced

Pipeline

Define your Pipeline using Groovy directly or pull it from source control.

Definition

Pipeline script



Script ?

```
1 pipeline {
2   agent any
3
4   stages {
5
6     stage('Build') {
7       steps {
8         bat 'javac src/main/java/TicketService.java'
9       }
10    }
11
12    stage('Test') {
13      steps {
14        bat 'javac src/main/java/TestRunner.java'
15        bat 'java -cp src/main/java TestRunner'
16      }
17    }
18  }
19 }
```

try sample Pipeline...

☒ Use Groovy Sandbox ?

Save

Apply

Activate Windows

Go to Settings to activate Windows.



Jenkins / TicketPipeline2 / #1

Status

✓ #1 (Apr 15, 2026, 6:33:49 PM)

Add description

Changes



Started by anonymous user

Console Output



No changes.

Edit Build Information

Delete build '#1'

Replay

Pipeline Steps

Workspaces

Next Build

Console

Started by user unknown or anonymous

[Pipeline] Start of Pipeline

[Pipeline] End of Pipeline

Finished: SUCCESS

```
PS C:\Users\Madhumitha\Desktop\TicketCancellationSystem> kubectl apply -f deployment.yaml
deployment.apps/ticket-app created
PS C:\Users\Madhumitha\Desktop\TicketCancellationSystem> kubectl apply -f service.yaml
The Service "ticket-service" is invalid: spec.ports[0].nodePort: Invalid value: 30007: provided port is already allocated
PS C:\Users\Madhumitha\Desktop\TicketCancellationSystem>
```

Activate Windows
Go to Settings to activate Windows.

Ln 13, Col 22 Spaces: 2 UTF-8 CRLF () YAML Go Live

Java: Ready

18:36:25 15-04-2026

```
PS C:\Users\Madhumitha\Desktop\TicketCancellationSystem> docker build -t ticket-system .
[+] Building 1.8s (9/9) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 189B
=> [internal] load metadata for docker.io/library/eclipse-temurin:17-jdk
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [1/4] FROM docker.io/library/eclipse-temurin:17-jdk@sha256:cc8d306395be4a8375277887d83e9e77eae05c26b6eba8ef320e5f0640e3a1fe
=> => resolve docker.io/library/eclipse-temurin:17-jdk@sha256:cc8d306395be4a8375277887d83e9e77eae05c26b6eba8ef320e5f0640e3a1fe
=> [internal] load build context
=> => transferring context: 5.54kB
=> CACHED [2/4] WORKDIR /app
=> [3/4] COPY src/main/java/ /app/
=> [4/4] RUN javac TicketService.java TestRunner.java
=> exporting to image
=> => exporting layers
=> => exporting manifest sha256:c272c742881637b78ff03c2da1c3aea6fd21f354db4bd25548a3a681f01397d6
=> => exporting config sha256:133abe50df76a43e3a36a1f4c17885b9ace0e02b3e6af2be7c1781cbad17b1b4
=> => exporting attestation manifest sha256:3907f39af22936a22e2e54c778bcabedf66106ae6b10e9004677f978ae8a7b19
=> => exporting manifest list sha256:1e2e34a016d5410b405103179211ee3f4525f567ffd064d54301a0a2f1938e06
=> => naming to docker-desktop://dashboard/build/desktop-linux/desktop-linux/wfgkmj08guos34ena8fsrn4nh (ctrl + click)
=> => unpacking to docker-desktop://dashboard/build/desktop-linux/desktop-linux/wfgkmj08guos34ena8fsrn4nh (ctrl + click)

View build details: docker-desktop://dashboard/build/desktop-linux/desktop-linux/wfgkmj08guos34ena8fsrn4nh
PS C:\Users\Madhumitha\Desktop\TicketCancellationSystem> docker run ticket-system
TestFullRefund: PASSED
TestHalfRefund: PASSED
TestNoRefund: PASSED
TestInvalidInput: PASSED

Total Tests: 4
Passed: 4
Failed: 0
PS C:\Users\Madhumitha\Desktop\TicketCancellationSystem> 
```

	ticket-system	latest	46c2a614c092	10 minutes ago	635.98 MB				
--	---------------	--------	--------------	----------------	-----------	--	--	--	--

PERSONAL

Search

Ctrl+K

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3.05 GB / 4.5 GB in use

8 images

Last refresh: 7 hours ago

Search

	Name	Tag	Image ID	Created	Size	Actions
☐	fund-transfer	latest	60fc451c0892	4 days ago	1.6 GB	
☐	login-app	latest	8e512bf8a319	7 days ago	635.99 MB	
☐	eclipse-temurin	17-jdk	cc8d306395be	8 days ago	639.46 MB	
☐	kindest/node	v1.34.3	08497ee19eac	4 months ago	1.35 GB	
☐	docker/desktop-cloud-provider-k8s	v0.5.0	4ad59ce20658	4 months ago	595.06 MB	
☐	envoyproxy/envoy	v1.36.4	08acc53db6f4	4 months ago	247.74 MB	
☐	docker/desktop-containerd-registry	v0.0.3	179a9ea3ae47	4 months ago	50.32 MB	
☐	ticket-system	latest	46c2a614c092	30 minutes ago	635.98 MB	

Engine running

Kubernetes running

RAM 3.09 GB

CPU 1.09%

Disk: 8.45 GB used (limit 1006.85 GB)

Terminal

Update available

Activate Windows

Go to Settings to activate Windows.

Showing 8 items

 Gordon BETA

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[Containers](#) / wizardly_chandrasekhar



wizardly_chandrasekhar

 cc06bfadffa9  [ticket-system:latest](#)

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[Stats](#)

TestFullRefund: PASSED
TestHalfRefund: PASSED
TestNoRefund: PASSED
TestInvalidInput: PASSED

Total Tests: 4
Passed: 4
Failed: 0